

Magic: the Gathering cEDH AI

Competitive Elder Dragon Highlander
(cEDH) · Multiplayer · Imperfect Information
Richard Lam · 2026

Preface

Why MTG is a harder AI problem than Chess or Go

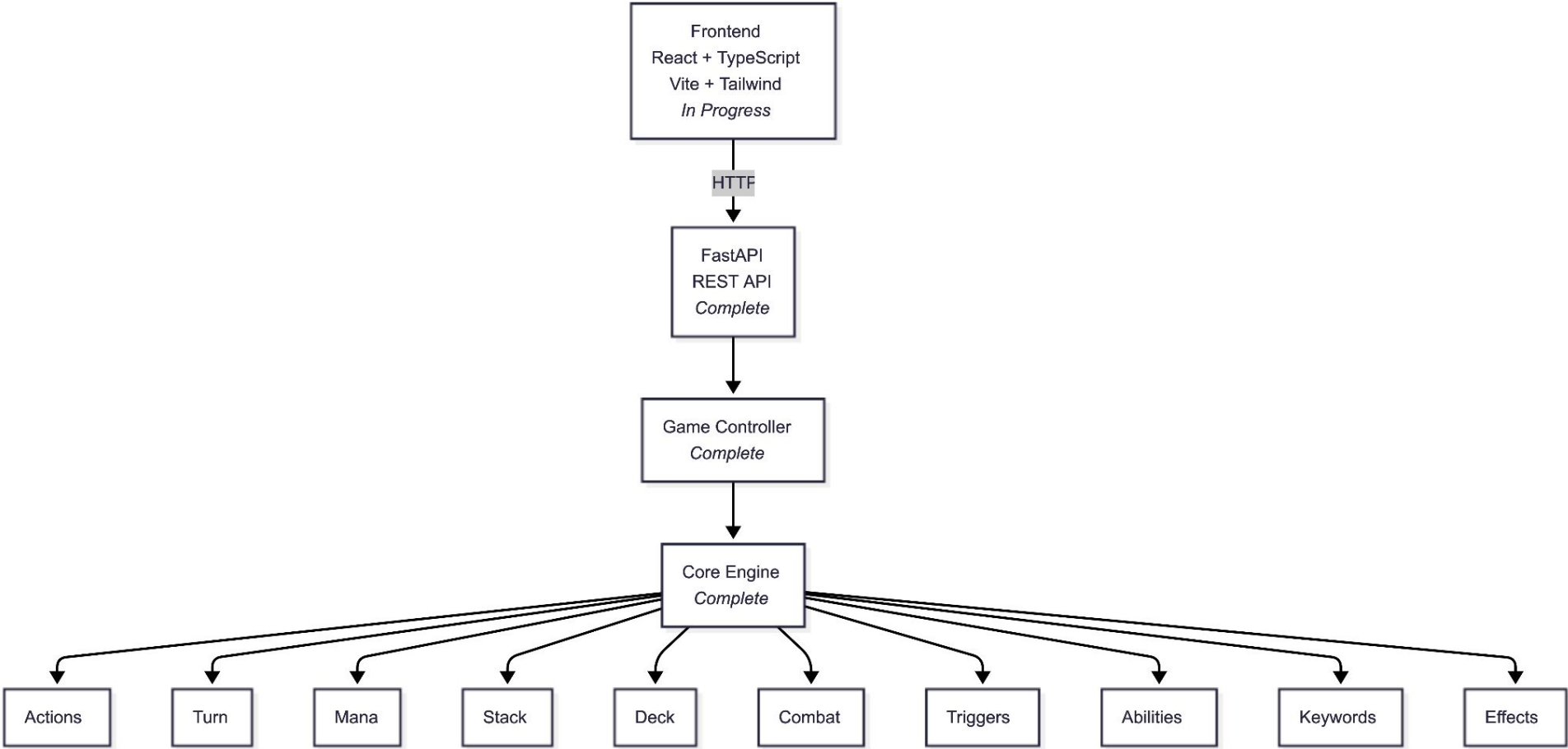
- MTG combines hidden information, multiplayer dynamics, and a complex rules engine; making it closer to a multi-agent system than a board game.

TABLE I
COMPARATIVE ANALYSIS OF CHESS, GO, AND MAGIC: THE GATHERING (MTG)

Aspect	Chess	Go	Magic: The Gathering (MTG)
Type	Abstract, perfect-information, two-player board game	Abstract, perfect-information, two-player board game	Collectible card game, multiplayer, imperfect information
Origin	6th century India / Europe evolution	China, 2000 BCE	1993, USA
Complexity	High tactical & positional depth, but finite	Extremely high; simple rules, vast strategy space	Extremely high; combinatorial depth, rules interactions, and emergent strategies
Game State Size	$\sim 10^{47}$ positions	$\sim 10^{170}$ positions	Effectively unbounded due to card combinations & interactions
Information	Perfect	Perfect	Imperfect (hidden cards, draws, opponent strategy)
Decision Type	Deterministic, tactical + strategic planning	Deterministic, long-term strategic planning	Probabilistic, adaptive, multi-layered strategic reasoning
Time to Master	Decades to become grandmaster	Lifelong mastery possible; very high ceiling	Lifelong mastery; constantly evolving meta, high variance
Recognition	Globally established, competitive tournaments	Culturally iconic in East Asia, growing worldwide	Recognized by MIT Technology Review as the world's most complex game; evolving competitive scene
Skills Required	Calculation, pattern recognition, strategy, endgame mastery	Pattern recognition, long-term planning, balance of influence	Systems thinking, probability, optimization, meta-analysis, adaptability, creativity

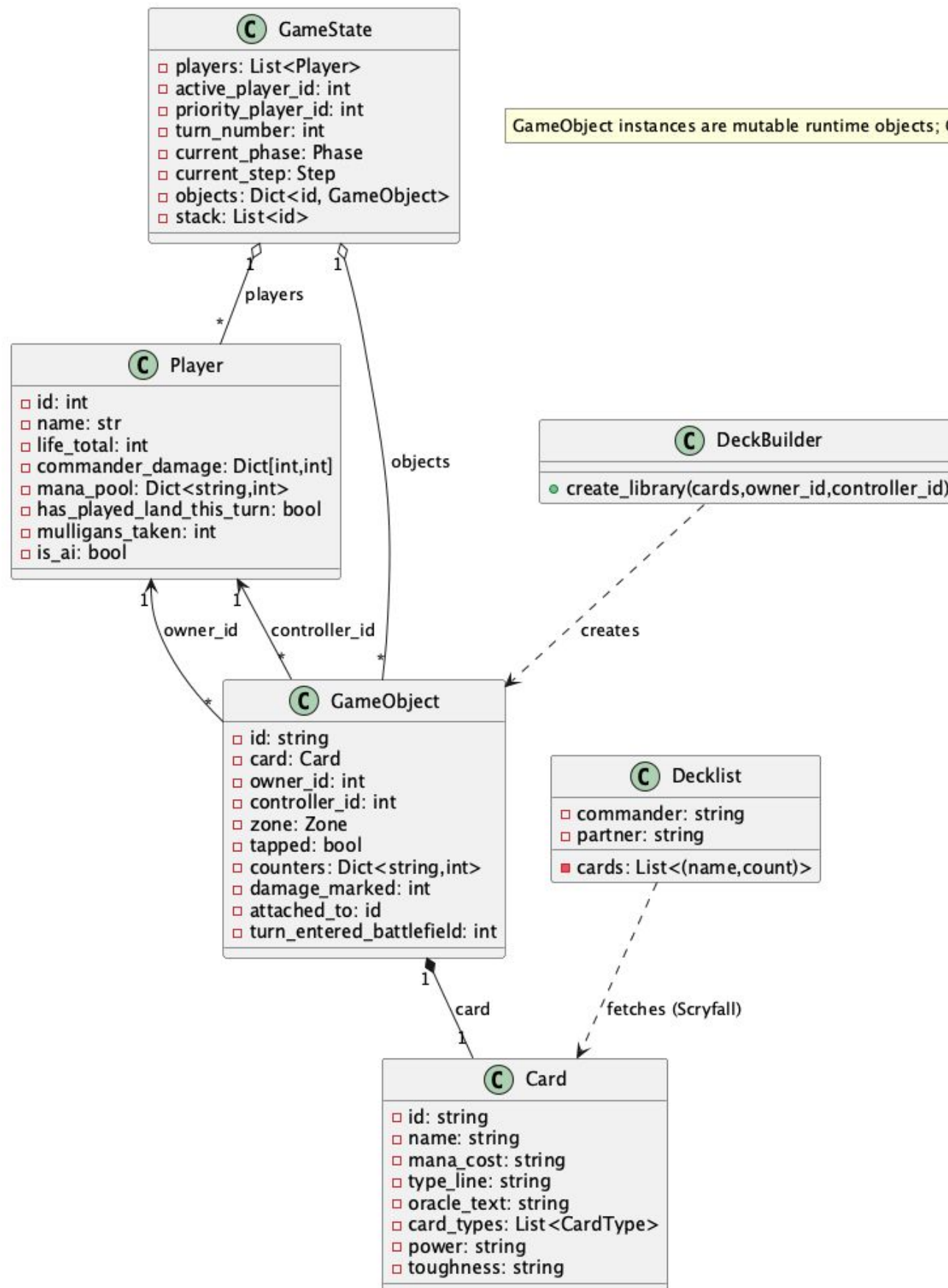
Software Architecture

UI → API → Game Controller → Core Rules Engine



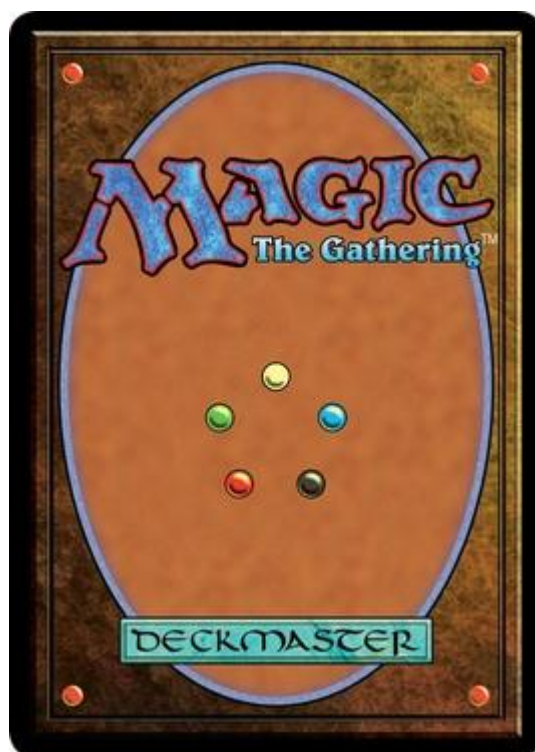
Game State / Data Model

Project cEDH – Data Model Overview



Core Engine

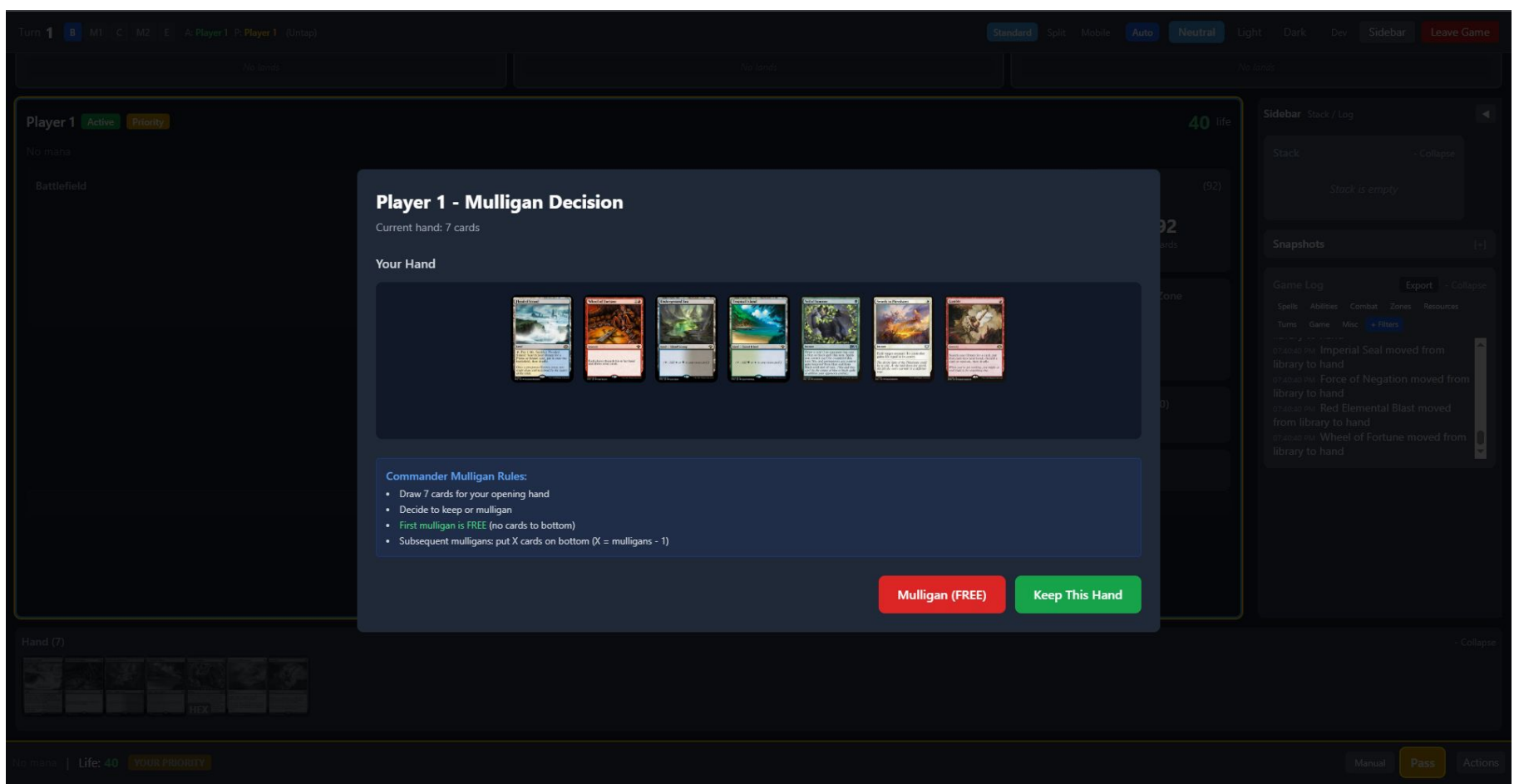
- Turn & priority system
- Stack resolution & counterspells
- Mana costs (hybrid, phyrexian, X)
- Combat (multiplayer attackers / blockers)
- Triggers, abilities, keywords
- Oracle text execution



End-to-End Gameplay

Work in Progress

- ✓ Create game from decklists
- ✓ Commander London Mulligan
- ✓ Full turn cycle
- ✓ Cast spells & interact on stack
- ✓ Combat, triggers, state-based actions
- ✓ AI players auto-pass priority



What's Next?

- Perfect backend mechanics
- Finish frontend polish
- Smarter AI (beyond auto-pass)
- Heuristic evaluation
- Monte Carlo Tree Search
- Self-play infrastructure
- Demo

Demo

CEDH AI

A DETERMINISTIC MAGIC: THE GATHERING RULES ENGINE AND SIMULATION PLATFORM
FOR COMPETITIVE COMMANDER RESEARCH, AI SELF-PLAY, AND PRIVATE PLAYTESTING.

ABOUT

PLAY

RESEARCH

LINKS

© CEDH AI. TEMPLATE: [HTML5 UP](#).

PLAY

CREATE GAME LOBBY

Set up a multiplayer session. Share the invite link — each player joins and submits their own decklist before the game starts.

PLAYERS

2 Players

SLOTS

SLOT 1: HUMAN

SLOT 2: AI

CREATE LOBBY

QUICK GAME

Play immediately against three AI opponents using the default Kenrith vs Kalamax vs Tymna/Thrasios vs Rograkh/Silas 4-player configuration. No setup required.

▶ PLAY VS AI

Demo continued

Turn 1B M1 C M2 E A: Player 1 P: Player 1 (Untap)

StandardSplitMobileAutoNeutralLightDarkDevSidebarLeave Game

No landsNo landsNo lands

Player 1ActivePriority

No mana

Battlefield

40 life

(92)

92 cards

Zone

(0)

Stack

Collapse

Stack is empty

Snapshots

[+]

Game Log

Export

Collapse

SpellsAbilitiesCombatZonesResources

TurnsGameMiscFilter

Impetal Blast moved from library to hand

Force of Negation moved from library to hand

Red Elemental Blast moved from library to hand

Wheel of Fortune moved from library to hand

Player 1 - Mulligan Decision

Current hand: 7 cards

Your Hand

Commander Mulligan Rules:

- Draw 7 cards for your opening hand
- Decide to keep or mulligan
- First mulligan is FREE (no cards to bottom)
- Subsequent mulligans: put X cards on bottom (X = mulligans - 1)

Mulligan (FREE)Keep This Hand

Hand (7)

HEX

No mana | Life: 40YOUR PRIORITY

ManualPassActions

Turn 1B M1 C M2 E A: Player 1 P: Player 1

StandardSplitMobileAutoNeutralLightDarkDevSidebarLeave Game

No landsNo landsNo lands

Player 1ActivePriority

No mana

Battlefield

40 life

(92)

92 cards

Library

Command Zone

Graveyard

Exile (0)

Stack

Collapse

Stack is empty

Snapshots

[+]

Game Log

Export

Collapse

SpellsAbilitiesCombatZonesResources

Filters

Impetal Blast moved

Fortune moved from

Phase — upkeep

Phase — draw

Main Phase — main

Kenrith, the Returned King

Legendary Creature — Human Noble

- All creatures gain trample and haste until end of turn.
- Put a +1/+1 counter on target creature.
- Target player gains 5 life.
- Target player draws a card.
- Put target creature card from a graveyard onto the battlefield under its owner's control.

5/5

Hand (7)

HEX

No mana | Life: 40YOUR PRIORITY

ManualEndPassActions